

Selection Sort - Viva Preparation

This document contains viva preparation notes, important concepts, functions used, algorithm explanation, and expected viva questions for Selection Sort implemented in Java.

1. Important Concepts

Concept	Explanation
Selection Sort	A sorting algorithm that repeatedly selects the minimum element and places it at its correct position.
In-place Sorting	Sorting done without extra array.
Comparison Sort	Sorting performed using comparisons.
Time Complexity	$O(n^2)$
Space Complexity	$O(1)$

2. Selection Sort Algorithm

1. Start from first element.
2. Find minimum element in remaining array.
3. Swap minimum element with current position.
4. Repeat until array becomes sorted.

3. Functions Used in Code

Function	Purpose
selectionSort()	Performs Selection Sort on array.
printArray()	Prints array elements.
main()	Takes input and calls sorting function.

4. Dry Run Example

Array: 5 3 1 4 2

Pass 1 → 1 3 5 4 2

Pass 2 → 1 2 5 4 3

Pass 3 → 1 2 3 4 5

Sorted Array → 1 2 3 4 5

5. Important Viva Questions

- What is Selection Sort?
- How does Selection Sort work?
- What is time complexity of Selection Sort?
- What is space complexity?

- Why is Selection Sort called in-place sorting?
- Difference between Bubble Sort and Selection Sort?
- Is Selection Sort stable?
- How many swaps occur in Selection Sort?
- What is best case time complexity?
- What is worst case time complexity?

6. Short Viva Answers

Question	Answer
What is Selection Sort?	A sorting algorithm that selects minimum element repeatedly.
Time Complexity?	$O(n^2)$
Space Complexity?	$O(1)$
Stable or Not?	Not stable by default.
Why in-place?	No extra array is used.

7. Complexity Analysis

Best Case Time Complexity: $O(n^2)$
 Average Case Time Complexity: $O(n^2)$
 Worst Case Time Complexity: $O(n^2)$
 Space Complexity: $O(1)$